

RAHUL POTHIRENDI

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Education

Northeastern University

Master of Science in Software Engineering Systems

Dec 2025

Boston, MA

Technical Skills

Languages: Python, Java, JavaScript/TypeScript, C++, Go, SQL

ML/AI: PyTorch, TensorFlow, LangChain, LangGraph, RAG, Vector Databases (Pinecone, Chroma)

Frontend: React.js, Next.js, TypeScript, Redux, HTML/CSS, Figma

Backend: Node.js, Express, FastAPI, Flask, Spring Boot, REST APIs, Microservices, Firebase (Firestore, Cloud Functions)

Cloud & Infrastructure: AWS (ECS, S3, EC2, RDS, SageMaker), GCP, Azure, Firebase, Docker, Kubernetes, Terraform

Professional Experience

IpserLabs

Jan 2025 – May 2025

Software Engineer Intern

Boston, MA

- Developed and integrated **Java Spring Boot** backend APIs with **React.js** components for the UncoverGem quiz platform, enabling seamless data flow between frontend and backend and supporting thousands of concurrent users with low latency and responsive UI performance.
- Partnered with product managers, frontend developers, and QA engineers to design and implement RESTful services, automate testing workflows, and enhance accessibility across application modules, ensuring consistent quality, improved performance, and seamless user experience across the platform.
- Engineered **React/TypeScript** UI with **Redux** state management, improving page load speed by 35% through code-splitting and lazy loading, achieving 95+ Lighthouse scores across devices.

Aditya University

Jan 2022 – Aug 2023

AI Full Stack Engineer

Remote

- Architected **FastAPI** microservices processing 10K+ concurrent requests with sub-200ms P99 latency, serving real-time academic data to 10,000+ users with 99.9% uptime through horizontal scaling and caching.
- Implemented distributed **Kafka** streaming pipeline handling 500K+ events/day with 60% throughput improvement; launched cross-platform **Flutter** mobile app deployed to 8,000+ students with 70% DAU, featuring offline-first architecture and Firebase push notifications, reducing server load by 45%.
- Developed multimodal sentiment analysis models combining audio (custom **CNN**), text (**BERT**), and image (**ResNet**) inputs for emotion recognition and sentiment classification; built AI-based proctoring solution improving exam integrity by 30%.
- Enhanced document digitization pipeline (Doc2Scan) using advanced **computer vision** and face recognition with optimized image preprocessing and feature extraction for higher accuracy.

Technical Projects

LearnLab - Agentic AI Learning System | *LangGraph, RAG, Gemini LearnLM, GCP, Terraform*

Jan 2026

- Architected AI-driven educational platform using **LangGraph** and RAG with Gemini LearnLM, transforming static academic PDFs into interactive learning content (two-person podcasts, flashcards, quizzes, blogs) through multi-agent orchestration enabling adaptive, human-AI collaborative learning experiences.
- Deployed scalable infrastructure on **GCP** using **Terraform** IaC with automated CI/CD via GitHub Actions, achieving zero-downtime deployments and seamless content generation workflows for personalized learning paths based on student progress.

QuestionCraft - AI Interview Platform | *LangChain, LangGraph, FastAPI, GPT-4, PostgreSQL*

Jan 2026

- Designed multi-agent AI system using **LangChain** and **LangGraph**, orchestrating Resume Analyzer, Question Generator, and Solution Validator agents with GPT-4, generating personalized interview questions in <30 seconds by analyzing candidate resumes (PDF/DOCX with spaCy NLP) and matching company tech stack requirements.
- Implemented backend API with async SQLAlchemy ORM and PostgreSQL achieving <200ms response time, integrated Judge0 code execution sandbox for real-time interview sessions with automated test case validation supporting Python, JavaScript, and Java with secure containerized execution and <2s latency.

Production MLOps Pipeline on AWS | *Python, Docker, Terraform, AWS ECS, React*

Sep 2025

- Engineered end-to-end ML inference system deploying YOLO and DepthAnything V2 models for traffic analysis, processing 30 FPS video streams with <100ms latency on containerized Flask APIs, serving real-time predictions to React dashboard via **WebSocket** updates.
- Provisioned AWS infrastructure using **Terraform** with **ECS Fargate** Auto Scaling (2-10 tasks), ALB path-based routing, and multi-AZ deployment achieving 99.95% availability; optimized CI/CD with AWS CodeBuild, compressing container images by 40% and reducing cold start by 60%.